Netflix Titles Analysis

Project documentation: data cleaning, SQL business questions, Python analysis and Power BI dashboard

Prepared 1 November 2026

8,790	6,126	2,664	2008 to 2021
titles in the cleaned dataset	movies (69.7%)	TV shows (30.3%)	years titles were added

1. Project overview

This project analyses a catalogue of Netflix movies and TV shows from end to end. The raw data was cleaned with Python (pandas), business questions were answered with SQL in MySQL Workbench, the data was explored further in a Jupyter notebook, and the results were presented in a Power BI dashboard. This document records what was done at each step, the results, and the points that need attention.

File	Tool	Purpose
Dataset.csv	Python (pandas)	The cleaned dataset: 8,790 rows and 10 columns.
finalout.sql	MySQL Workbench	Seven business questions, run on database *auspify*, table *netfilx*.
data_analysis.ipynb	Python (pandas, matplotlib)	Exploratory analysis in nine sections, including a check of the SQL results.
out1.pbix	Power BI	Dashboard *Global Streaming Media & Production Insights*.

Workflow: raw data, then cleaning in pandas, then (a) SQL business questions, (b) notebook analysis and (c) the Power BI dashboard, with the SQL results cross-checked against pandas.

2. The dataset

Each row is one title. The file has 10 columns and 8,790 rows, with no duplicate *show_id* values.

Column	Type	Description
show_id	text	Unique ID such as s1 or s14.
type	text	*Movie* (6,126) or *TV Show* (2,664).
title	text	Title name.
director	text	One or more names separated by commas. *Not Given* when unknown (2,588 titles, 29.4%).
country	text	One country per title. *Not Given* for 287 titles (3.3%). 86 distinct values including *Not Given*.
date_added	text (M/D/YYYY)	Date added to Netflix, from 1 Jan 2008 to 25 Sep 2021.

Column	Type	Description
release_year	integer	Year of release, from 1925 to 2021.
rating	text	14 distinct ratings, for example TV-MA, TV-14, R.
duration	text	Minutes for movies (*90 min*) and seasons for TV shows (*1 Season*).
listed_in	text	Genre tags separated by commas, for example *Dramas, International Movies*.

3. Data cleaning with pandas

The cleaning code was not part of the files provided for this documentation. This section therefore records the **state of the cleaned data**, checked directly on Dataset.csv. Add the exact cleaning steps (for example the columns dropped and the way missing values were filled) next to this table.

Check on the cleaned file	Result
Rows and columns	8,790 rows, 10 columns
Missing values (blank cells)	0 in every column
Duplicate rows or duplicate show_id	0
Missing directors	Filled with the placeholder *Not Given*: 2,588 titles (29.4%)
Missing countries	Filled with the placeholder *Not Given*: 287 titles (3.3%)
Dates	All 8,790 values parse as M/D/YYYY. Stored as text, converted to dates in the analysis.
Duration	Always a number plus a unit (min, Season or Seasons). The number is extracted in the analysis.
Countries per title	Exactly one (no title lists several countries)

Derived columns created inside the analysis notebook:

- **year_added** and **month_added**, taken from date_added.
- **duration_num**, the number of minutes (movies) or seasons (TV shows).
- **genre**, one row per genre tag, made by splitting listed_in.
- **audience**, a grouping of ratings into Kids, Teens, Adults and Unrated.
- **age_at_addition**, year added minus release year.

Known data issue: country = Pakistan. Pakistan ranks 4th with 421 titles, but 402 of them (95%) have no director, against 26% for all other titles, and 83% are TV shows. Many of these rows are clearly not Pakistani productions, for example *The Great British Baking Show: Holidays*, *Chef's Table: France*, *Barney and Friends* and *Surviving Escobar - Alias JJ*. The likely cause is a country-filling step during cleaning. Pakistan figures in the SQL results, the notebook and the dashboard should be treated as unreliable until this is reviewed. The top three countries are not affected.

4. Business questions in SQL

The questions were answered in MySQL Workbench on database *auspify*, table *netfilx*. Every query below was re-run in pandas, and every result matches the MySQL output (section 5.5).

Q1. How many movies and TV shows?

```
Select type,count(*) as "Count" from netfilx
group by 1
```

type	Count
Movie	6,126
TV Show	2,664

Q2. Year-wise count (release year, newest first)

```
select release_year, count(*) AS "Count" from netfilx
group by 1
order by 1 desc
```

release_year	Count
2021	592
2020	953
2019	1,030
2018	1,146
2017	1,030

74 rows returned (74 distinct release years). The five newest years are shown. 2021 is a partial year.

Q3. Top 3 release years

```
select release_year, count(*) AS "Count" from netfilx
group by 1
order by 2 desc
limit 3
```

release_year	Count
2018	1,146
2017	1,030
2019	1,030

2017 and 2019 are tied at 1,030. The next year (2020, 953) is lower, so the top 3 is not affected by the tie.

Q4. Top 3 countries by number of titles

```
select country,count(*)as"Count" from netfilx
group by 1
order by 2 desc
limit 3
```

country	Count
United States	3,240
India	1,057
United Kingdom	638

Q5. Top 3 directors by country (all titles)

```
Select country,director, count(*)as "Count" from netfilx
where director !='Unknown Director'
group by 1,2
order by 3 desc
limit 3
```

country	director	Count
India	Rajiv Chilaka	20
United Kingdom	Alastair Fothergill	18
India	Suhas Kadav	16

This returns the three highest (country, director) pairs overall, not the top three inside each country.

Q6. Top 3 TV show directors by country

```
Select country,director, count(*)as "Count" from netfilx
where director !='Unknown Director' and  type !='Movie'
group by 1,2
order by 3 desc
limit 3
```

country	director	Count
United Kingdom	Alastair Fothergill	14
United States	Mark Thornton, Todd Kauffman	4
United States	Stan Lathan	3

Q7. Top 3 movie directors by country

```
Select country,director, count(*)as "Count" from netfilx
where director !='Unknown Director' and  type !='TV Show'
group by 1,2
order by 3 desc
limit 3
```

country	director	Count
India	Rajiv Chilaka	19
India	Suhas Kadav	16
United States	Marcus Raboy	15

Notes on the SQL

- **Missing-director placeholder.** The queries filter `director != 'Unknown Director'`, but Dataset.csv marks missing directors as *Not Given*. The MySQL table must have used *Unknown Director*. If Dataset.csv is loaded into MySQL, change the filter to `'Not Given'`; otherwise the top row of Q5 becomes *United States / Not Given (839)*.
- **Tie at third place in Q5.** Suhas Kadav (India) and Marcus Raboy (United States) both have 16 titles, but `limit 3` returns only one of them and MySQL does not guarantee which. To keep both, rank with `DENSE_RANK() OVER (ORDER BY Count DESC)` (MySQL 8+) and keep ranks 1 to 3.
- **Co-directed titles.** The SQL groups on the raw director text, so *Mark Thornton, Todd Kauffman* counts as one "director". The notebook splits co-directors, which is why Rajiv Chilaka has 20 titles in the SQL result (titles he directed alone, all in India) and 23 in the notebook (including 3 co-directed titles).
- **Running the whole file.** The file has no semicolons, and comments are written as `--text`. MySQL needs a space after the two dashes (`-- text`); without it the comment is read as SQL, which most likely caused the Error 1064 seen in the output panel. Adding semicolons and the space lets the file run in one go.

5. Analysis in the Python notebook

The notebook *data_analysis.ipynb* reads Dataset.csv, checks it, and works through nine sections. All charts use one colour code: **blue = movies** and **orange = TV shows**. Figures below are taken from the notebook run.

5.1 Catalogue mix and growth

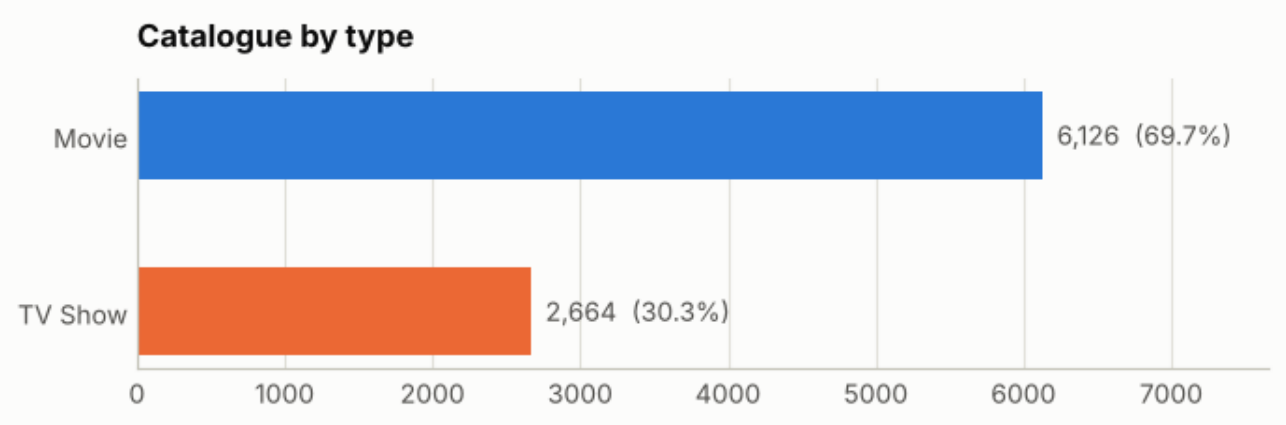


Figure 1. Movies make up 69.7% of the catalogue, TV shows 30.3%.

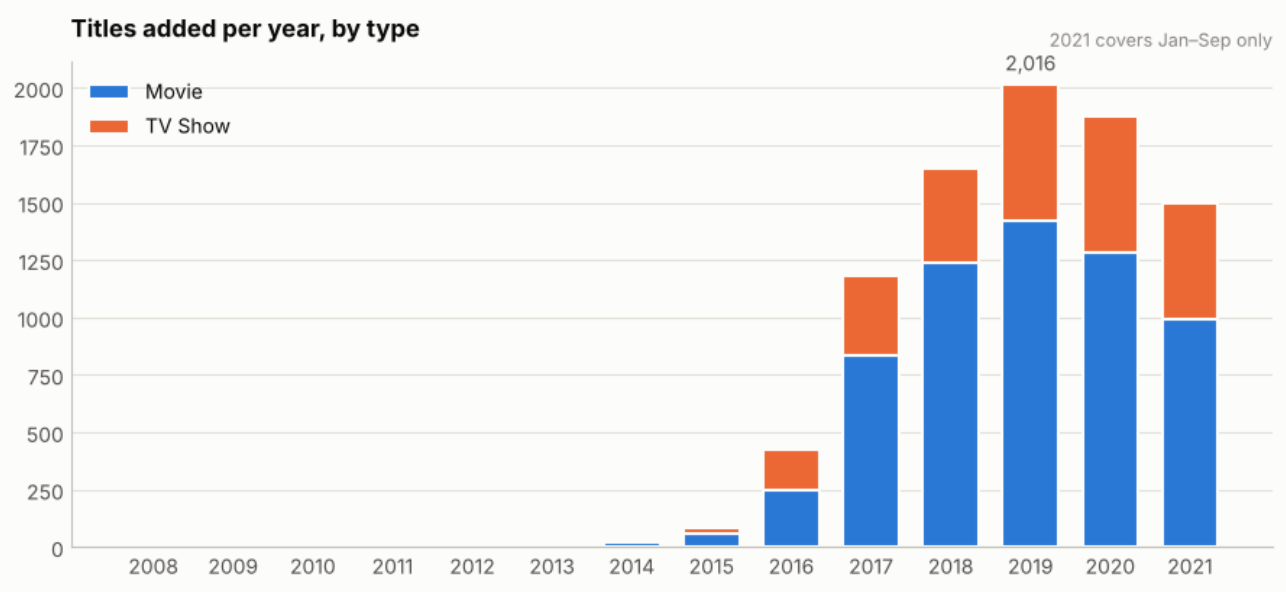


Figure 2. Titles added per year. Additions peaked in 2019 at 2,016 titles. 2021 covers January to September only.

Year added	Movies	TV shows	Total	TV share
2015	56	26	82	31.7%
2016	251	175	426	41.1%
2017	836	349	1,185	29.5%
2018	1,237	411	1,648	24.9%
2019	1,424	592	2,016	29.4%
2020	1,284	595	1,879	31.7%
2021 (to 25 Sep)	993	505	1,498	33.7%

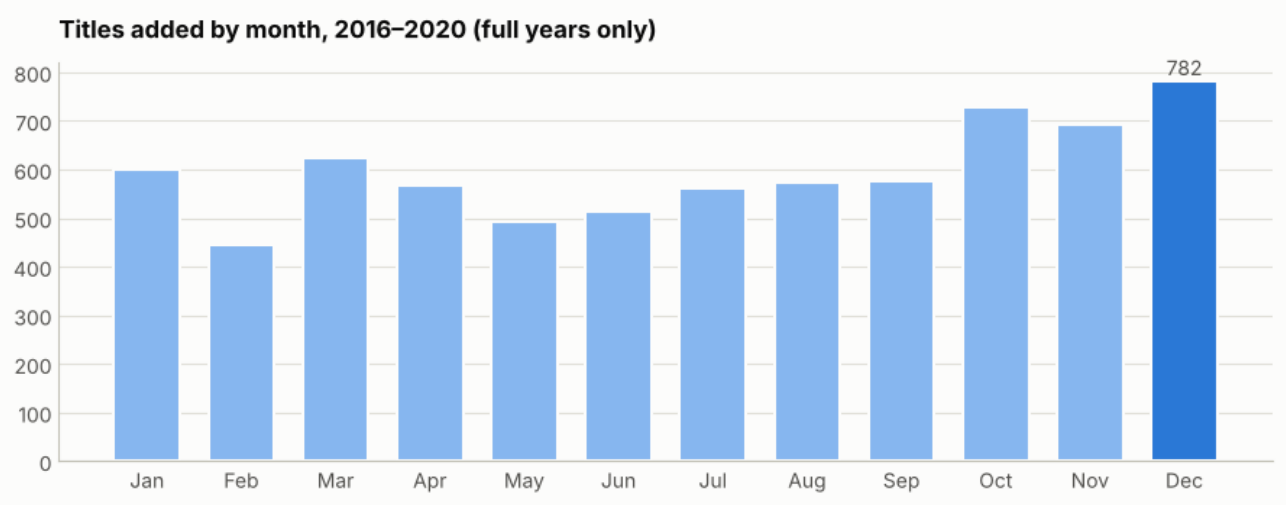


Figure 3. Titles added by month, 2016 to 2020 (full years only). December is the busiest month (782) and February the quietest (445).

5.2 Countries

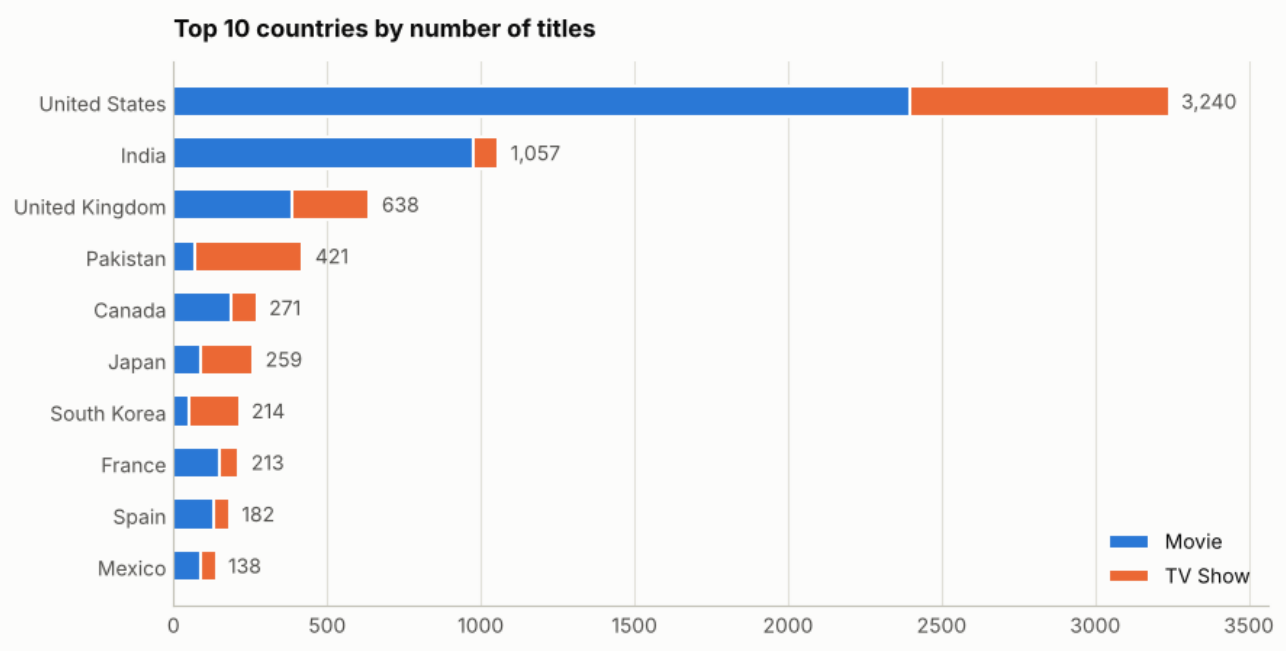


Figure 4. Top 10 countries. The United States leads with 3,240 titles, then India (1,057) and the United Kingdom (638). Titles with country Not Given are excluded.

Country	Movies	TV shows	Total	TV share
United States	2,395	845	3,240	26.1%
India	976	81	1,057	7.7%
United Kingdom	387	251	638	39.3%
Pakistan (see section 3)	71	350	421	83.1%
Canada	187	84	271	31.0%
Japan	87	172	259	66.4%
South Korea	49	165	214	77.1%
France	148	65	213	30.5%
Spain	129	53	182	29.1%
Mexico	90	48	138	34.8%

5.3 Genres, duration and ratings

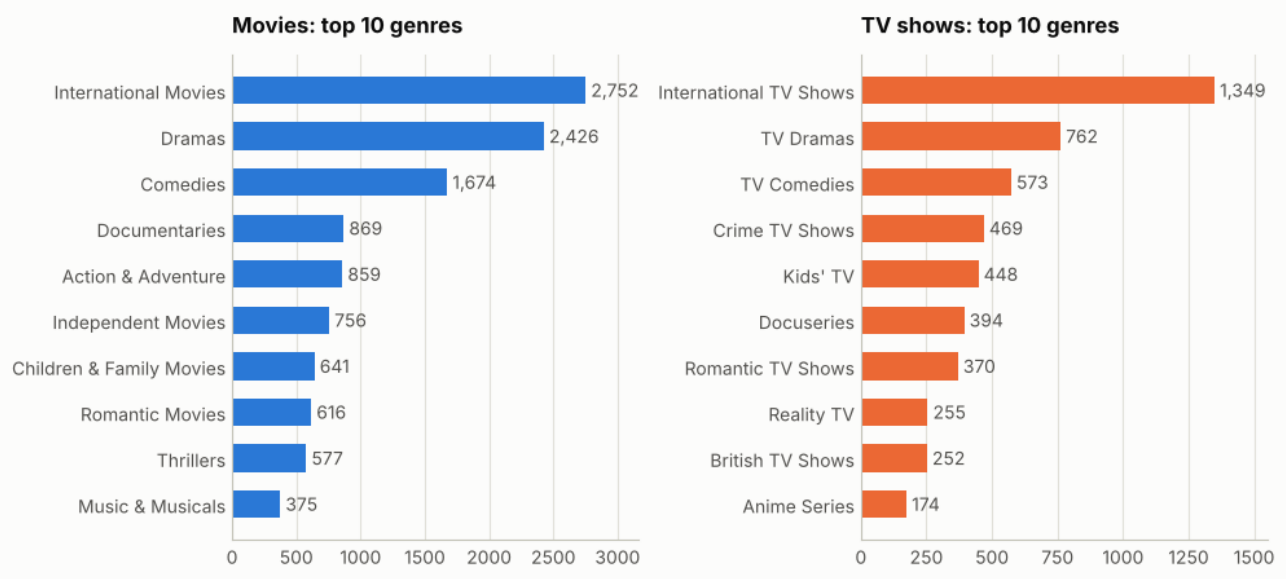


Figure 5. Top 10 genre tags. A title counts once in each of its genres, so genre counts add up to more than the number of titles.

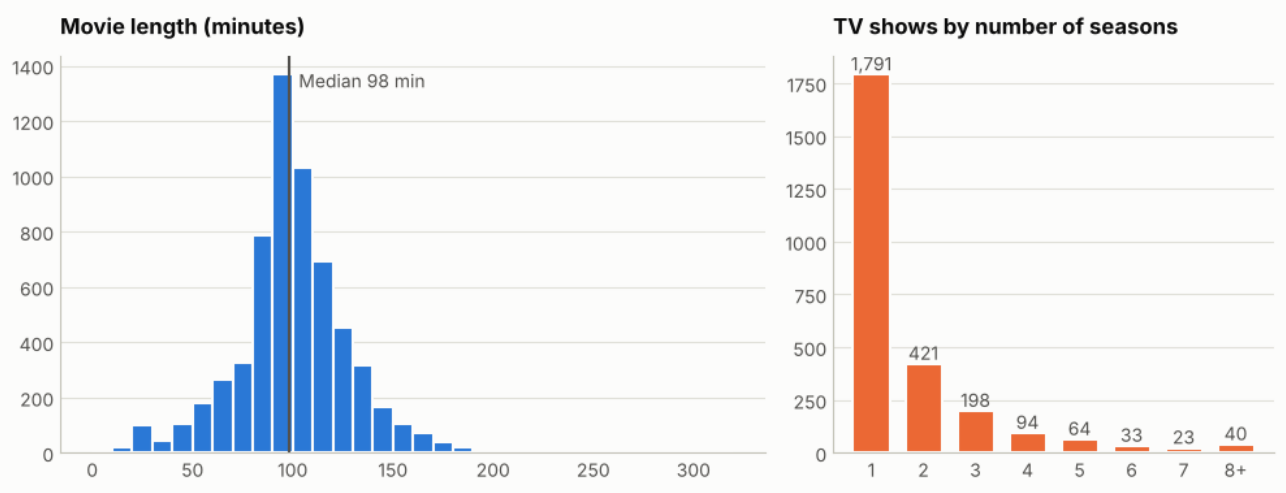


Figure 6. The median movie runs 98 minutes. 67% of TV shows have a single season and 83% have one or two seasons (the last bar groups shows with 8 or more seasons).

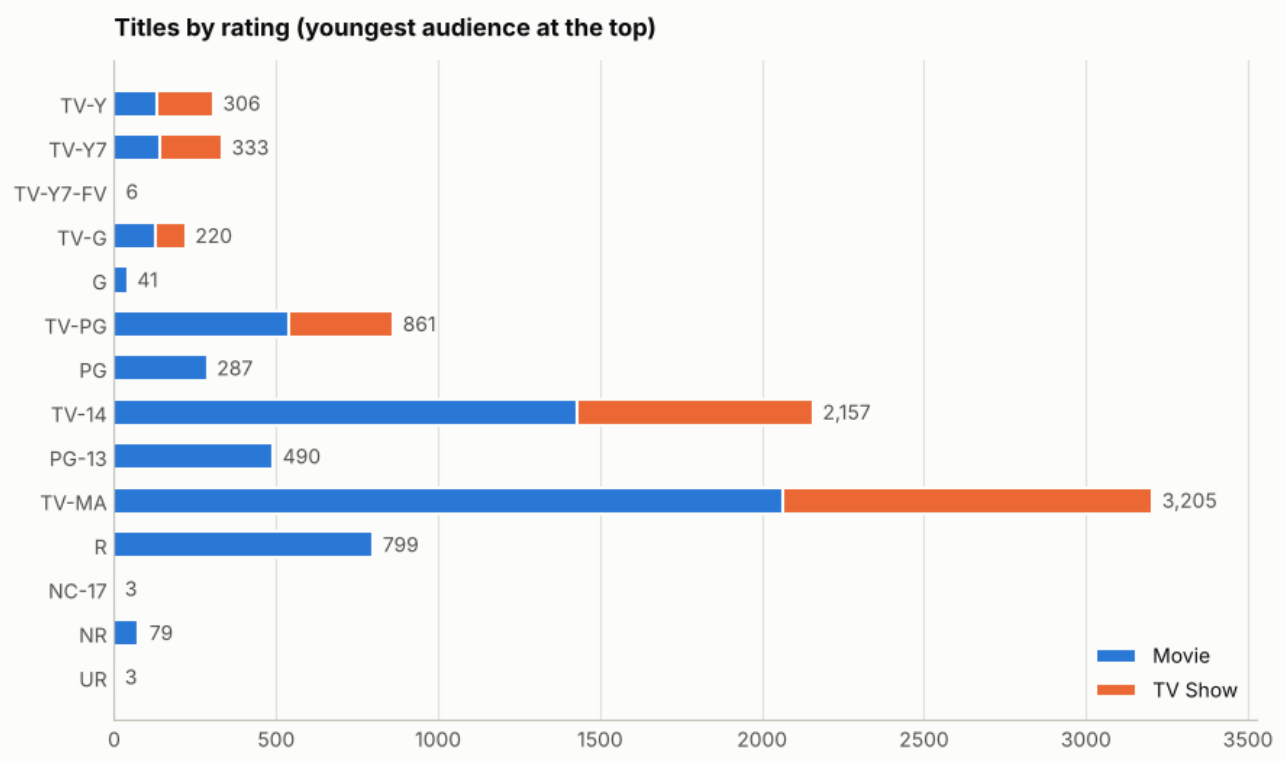


Figure 7. Titles by rating, youngest audience at the top. TV-MA (3,205) and TV-14 (2,157) are the most common ratings.

Audience group	Movies	TV shows
Kids (TV-Y, TV-Y7, TV-Y7-FV, TV-G, G)	7.2%	17.4%
Teens (TV-PG, PG, TV-14, PG-13)	44.8%	39.5%
Adults (TV-MA, R, NC-17)	46.7%	43.0%
Unrated (NR, UR)	1.3%	0.2%

Share of each type's titles by audience group. The grouping is a simplification of the rating scale.

5.4 Directors and content age

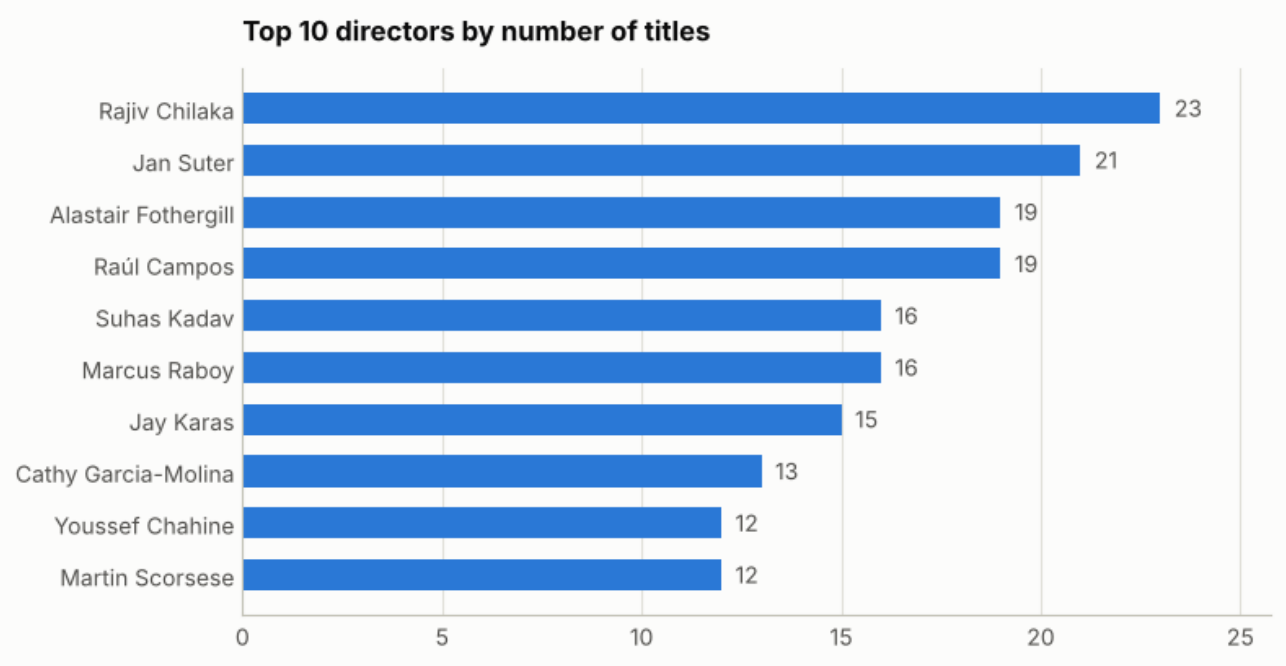


Figure 8. Top 10 directors among titles with a named director (71% of titles). Co-directors are split, so each person is counted separately. There are 4,991 distinct directors.

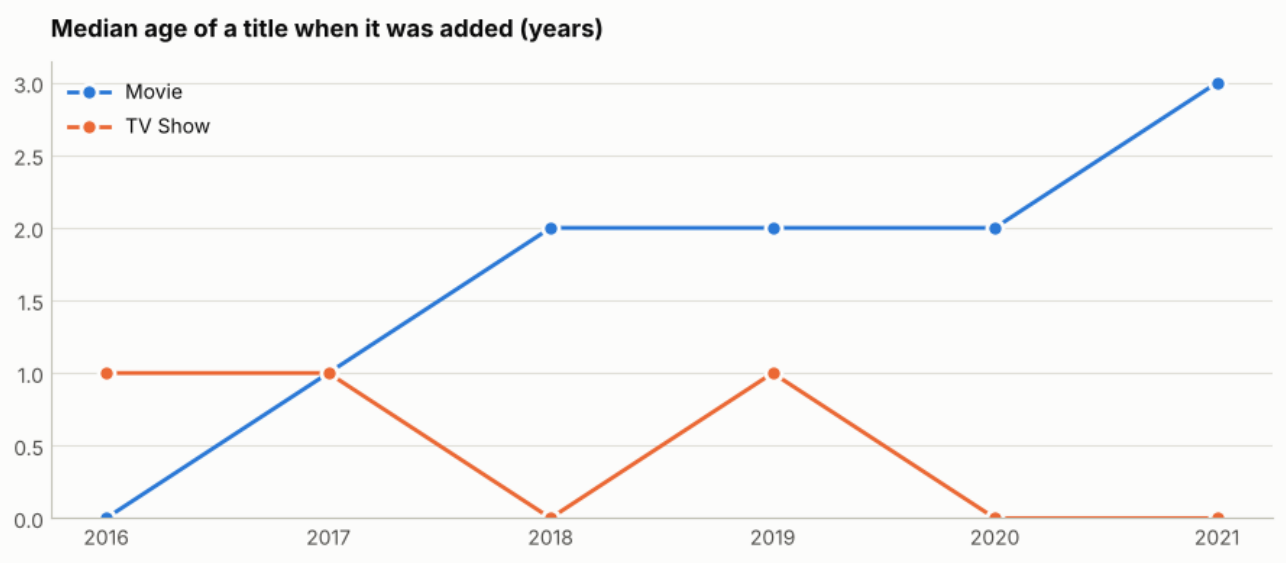


Figure 9. Median age of a title when it was added. Movies arrive later over time (0 years in 2016, 3 years in 2021) while TV shows arrive within a year of release. Overall, 63% of titles were added within two years of release, and 14 titles were added before their release year.

5.5 Cross-check of the SQL results

Section 9 of the notebook repeats all seven SQL questions in pandas and compares them with the MySQL numbers. All checks match: Q1 (type counts), Q2 (year-wise counts and the 74 distinct years), Q3 (top years), Q4 (top countries, and 86 distinct countries) and Q5 to Q7 (top directors). The notebook also lists the top three directors inside each of the top three countries, which is the other reading of "top 3 director by country".

Country	Top directors (titles)
India	Rajiv Chilaka (20), Suhas Kadav (16), David Dhawan (9)
United Kingdom	Alastair Fothergill (18), Edward Cotterill (4), Chris Howe (3)
United States	Marcus Raboy (16), Jay Karas (14), Jay Chapman (12)

6. Power BI dashboard

The dashboard *Global Streaming Media & Production Insights* (file out1.pbix) presents the same data as one interactive page. The .pbix file itself could not be opened for this document, so the description below is based on a screenshot of the page.

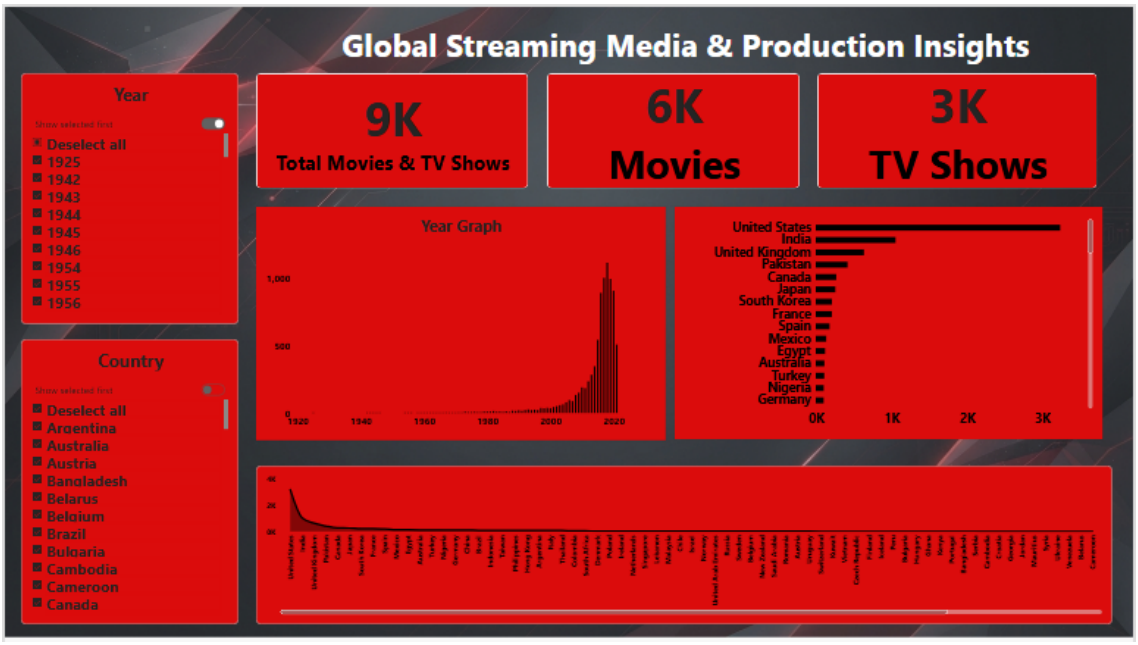


Figure 10. The dashboard page (screenshot).

Element	What it shows
Year slicer	Filters the whole page by release year (the list starts at 1925).
Country slicer	Filters the whole page by country.
Three KPI cards	Total movies and TV shows (9K), movies (6K) and TV shows (3K). These are rounded versions of 8,790, 6,126 and 2,664.
Year Graph	Column chart of titles by year, rising steeply to a peak in the late 2010s. It appears to use release year; the peak of just over 1,000 matches the SQL result for 2018 (1,146).
Country bar chart	Titles per country, led by United States, India and United Kingdom, then Pakistan, Canada, Japan, South Korea, France, Spain and Mexico.
Bottom column chart	Titles across all countries, showing a long tail after the top few countries.

The KPI values and the top countries agree with the notebook and SQL results. The Pakistan bar in the country chart carries the data issue described in section 3.

7. Key findings

- **Size and mix:** 8,790 titles: 6,126 movies (69.7%) and 2,664 TV shows (30.3%).
- **Growth:** additions rose from 82 in 2015 to 1,185 in 2017 and peaked at 2,016 in 2019. TV shows were 25% to 34% of additions in every year since 2017.
- **Timing:** December is the busiest month for additions (782 titles, 2016 to 2020) and February the quietest (445).
- **Countries:** the United States has 3,240 titles (38% of titles with a known country), India 1,057 and the United Kingdom 638. India is 92% movies, while South Korea (77%) and Japan (66%) lean towards TV shows.
- **Genres:** International Movies (2,752) and Dramas (2,426) lead among movies. International TV Shows (1,349) and TV Dramas (762) lead among TV shows.
- **Audience:** adult ratings cover 47% of movies and 43% of TV shows. Kids content is 17% of TV shows but only 7% of movies.
- **Duration:** the median movie runs 98 minutes, and 83% of TV shows have one or two seasons.
- **Directors:** 29% of titles have no director. Rajiv Chilaka has the most titles (23), then Jan Suter (21).
- **Content age:** 63% of titles were added within two years of release and 71% were released in 2015 or later.

8. Points to review and next steps

#	Point	Suggested action
1	Country *Pakistan* looks wrongly filled for about 400 titles (section 3).	Review the country-filling step in the cleaning code, re-export Dataset.csv, and refresh the SQL table, notebook and dashboard.
2	Missing directors are *Not Given* in Dataset.csv but *Unknown Director* in the SQL filter.	Use one placeholder everywhere, or filter on both values.
3	Q5 has a tie at third place (16 titles).	Rank with DENSE_RANK() or add a second ORDER BY column so the result is repeatable.
4	The SQL file does not run as a single script (no semicolons, no space after `--`).	Add semicolons and a space after the two dashes.
5	2021 is a partial year (to 25 Sep 2021), and year charts show it as a drop.	Label 2021 as partial in charts and the dashboard.
6	Dashboard KPI cards show 9K, 6K and 3K.	Show exact values (8,790, 6,126, 2,664). Also consider darker or lighter text on the red panels and larger country labels in the bottom chart, which are hard to read.

9. How to reproduce the work

Notebook

- Put *data_analysis.ipynb* and *Dataset.csv* in the same folder.
- Install the libraries: pandas, numpy and matplotlib.
- Open the notebook in Jupyter or VS Code and run all cells. Section 9 re-checks the SQL results.

SQL

- Create the database and import Dataset.csv into a table named *netfilx* (this spelling is used in the queries).
- Make sure the director placeholder matches the filter (see section 4), then run the queries in finalout.sql.

Dashboard

- Open out1.pbix in Power BI Desktop. Refresh the data after any change to Dataset.csv.